

I=0 PROCESSOR ERROR ROUTINES (IOPERR,3) 10/2/73 JGC/BPC

/CHANNEL 1 TAPE ERRORS (ALL IOT'S ARE RETURN 3)

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1HLT,      LAC I DCONT      /CHECK WHICH DEVICE
           SMA
           JMP I 1HLTF      /FASTRAND TOT CRASH ERROR
           LAC (200000)
1TERR,     LIO I FTAPE1     /GET PROPER FLAGS FOR CH1
           RIL 6S
           LFI
           LRG
TERR,      IOR (400000)
           DAC I ERCD01
TERR1,     LAW I 1
           ADD I PC
           LIA
           SZF I 6
           LAW IOPTSU
           SPI
           IOR (400000)
           DAC I PC
           CLA              /NOW SET FOR PROPER PC UNSTEP
           SZF I 1
           IDA              /WAS A THREE RETURN IOT
           SZF 6
           IDA              /WAS A TWO RETURN IOT
           CMA
           AAI
           DAC I TRAPPC
           JMP I SWAP

ILLSPC,    LAC (040000)     /ILLEGAL SPECIFICATION ERROR
           JMP 1TERR

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/CHANNEL 4 TAPE ERRORS (ALL IOT'S RETURN 2) FLAG 6 UNTOUCHED

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TAPERR,    0
           LAW TAPER1      /GET CHANNEL 1 TO ERROR ROUTINE
           DAP I FTAPE1
           JMP TPRLS2
TAPER1,    LAC ERWRD2
           DAC I ERCD02
           STF 1
           LAC TAPERR
           JMP TERR

ERWRD2,    0

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PAGE 2

THLT, LAC (200000 /TOTAL CRASH
JDA TAPERR

4LDPT, LAC (20000
IOR I UNPTR
AND (-100000
DAC I UNPTR
LAW 400
JDA TAPERR

4ILLSPC, LAC (040000)
JDA TAPERR

/CHANNEL 16 TAPE ERRORS

16ILLSPC, LAC (40000
IOR (400000
16TERR, DAC I RERCD1
LAC I RPC
LIA
DAC I RTRAPC
LAW I 1
ADD I (DCH 71)
SZF I 6
LAW IOPTSU
SPI
IOR (400000)
DAC I (DCH 71)
JMP I R0

NOTYOURS, LAC (020000
JMP 16ILLSPC+1

EMPTY, LAC (010000
JMP 16ILLSPC+1

WRING, LAW 200
JMP 16ILLSPC+1

F16ILLSPC, LAC (040000

JMP 16TERR

FNOTYOURS, LAC (020000

JMP 16TERR

FEMPTY, LAC (010000
JMP 16TERR

/END OF CHANNEL 1 SWAPPER

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SWAPE2,  AND CONTR2      /MTCU WANTS TO FREE A GUY
        ADD (3)
        SAD (TTBL 6)
        LAC (TTBL)
        SAS END2
        IOR B1
        DAC CONTR2
SWAPE7,  DZM ERSTART      /FORGET OLD ERRORS
        LAC I AC          /REPACK ACTIVE REGISTERS
        DAC I (IOR 50)
        LAC I IO
        DAC I (IOR 51)
        LAC I FLAGS
        DAC I (IOR 52)
        RCK
        LAI
        SUB I TIME
        SPA
        ADD (60000.)
        LIA
        TAD I IOPRT1
        DAC I IOPRT1
        SZL "U"SCF"U"CLL"U"CML
        IDX I IOPRT
        ISB 400           /SIGNAL CHANNEL 4
        LAC I ICORE       /LOOK UP CORE STATUS OF THIS GUY
        DAP CORSTAT
        LAC I ISTAT       /RESET CORE PTRS
        DAC I CORSTAT
        ISB 1700          /TELL SYSTEM SWAPPER
        DZM I HOTFLG
        DAP STATP
        LAI
        AND (37)
        SIR 5S
        NOP               /SIR OR SIL?
        DIO CLKCNT
        ADD I PRIORITY
        AND (7777)
        LIA
        AND (37)
        DAP I PRIORITY
        SIR 5S
        LAI
        ADD CLKCNT
        SZA I             /ANY CLOCK TICKS TO BE CHARGED?
        JMP I SWAPE5      /NO, SO SAVE TROUBLE AND RELEASE HIM NOW

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DAC CLKCNT
LAW 7777
AND I STATP
CMA
DAC RUNTM
LAC I PRIORITY
SAR 6S
SAR 6S
SZA I          /PROIRITY 0?
JMP LOOP       /YES, CHARGE TICKS*1
SPA
JMP SWAPE8     /NEG. PRI. GO CHARGE EXTRA TICKS
DAC SWAPE9     /POS. PRI. = HAS HE RUN LONG ENOUGH?
LAW 1
MUL CLKCNT
DIV SWAPE9
SWAPE9, 0
SWP
RAL 5S
ADD I PRIORITY /UPDATE FRACTION LEFT OVER
DAP I PRIORITY
SNI          /IS HE TO GET CHARGED ANYTHING?
JMP I SWAPE5  /NO
JMP LOOPA

SWAPE8, SCM          /THE LINK IS SET
MUL CLKCNT          /CALCULATE HOW BADLY THE GUY LOSES
RIR 1S
LOOPA, DIO CLKCNT

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LOOP,	LIO RUNTM	/FIND OUT HOW MANY CLOCK TICKS IT TOOK TO GET HERE
	FBC	
	LAI	
	XOR RUNTM	
	LIA	
	SAL 1S	
	ADD RUNTM	
	AAI	
	ADD CLKCNT	/ADD IN THE CURRENT CHARGES
	CMA	
	DAC SWAPE9	
	LIA	/NOW CONVERT BACK TO A QUEUE LEVEL
	FBC	
	LAI	
	XOR SWAPE9	
	SAL 1S	
	AAI	
	LIA	
	AND (777400)	
	SZA I	/WE PUTTING THE GUY ON LOW QUEUE?
	JMP LOOP1	/NO, SO HIS QUEUE IS OK
	AND (777000)	/DID THE GUY GO OFF THE END OF LOW QUEUE?
	SZA	
	JMP LOOP2	/YES, SO HE STAYS AT THE BOTTOM OF LOW QUEUE
	LAW 400	/GUY IS IN THE MIDDLE OF LOW QUEUE...
	AND RUNTM	/DID HE START IOT ON LOW QUEUE?
	SZA	
LOOP2,	LAW 777	/NO, SO HE MUST BEGIN AT THE BOTTOM
LOOP1,	IAI	
	AND (777)	
	DAP I STATP	/PUT UPDATED QUEUE BACK
	LAC STATP	
	SAS NETHND	/DID WE JUST CHARGE "NCP" ?
	JMP I SWAPE5	
	LAW 7777	
	AND I STATP	
	SUB (200)	
	SPA	
	JMP I SWAPE5	
	LAW 40	/GOOSE "NCP"
	DAP I STATP	
	JMP I SWAPE5	
STATP,	DCH .	
CLKCNT,	0	
RUNTM,	0	
CORSTAT,	DCH .	

/CHANNEL 1 ABNORMAL ROUTINE
 /CH1XE=0, NO PREVIOUS ERRORS UNRECOVERED
 /CH1XE=-0, PREVIOUS UNRECOVERED ERROR

ABNORMAL, CLA	/ARE THERE ANY PREVIOUS UNRECOVERED ERRORS
SAS I CH1XE	
JMP I CH1A	/YES=IGNORE ERROR STATUS
RRI	/GET STATUS OF CHANNEL
RIL 2S	/IS IT A DEVICE ERROR
SPI	
JMP DEVICE	
RIL 2S	/WAS DATA LATE
SPI	
JMP DATA	/RE=EXECUTE COMMANDS (HANDLE AS DATA ERROR)
RIL 8S	
LFI	/LOAD FLAGS WITH STATE OF COMPARISON BITS
RIL 2S	
SPI	
JMP 1HLT	/COMMAND REJECTED
SZF 5	
JMP .+4	
SZF 2	
SZF I 6	
JMP DATA	
CLC	/COMPARISON ERROR=RESET SWITCH
DAC I CH1XE	/SET AS PREVIOUS UNRECOVERED ERROR
LIO I CHNFLG	/RESTORE THE FLAGS
LAC ERSTART	/CHECK IF IN PROCESS OF PROCESSING AN ERROR
SZA	
LIO SCHFLG	/YES=GET FLAGS FROM SAVED REGISTER
LFI	
DZM ERSTART	
JMP I COMPX	/EXIT TO COMPARISON ERROR ROUTINE

/DEVICE PROBLEMS

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DEVICE,  RRI
        LAI
        AND (3           /WHICH CONTROLLER
        SAS (1
        JMP ATAPE        /TAPE ERROR
        RRI 100          /CHECK FOR A DEAD FASTRAND
        RIR 7S
        SPI
        HLT              /UNIT UNAVAILABLE - HUH?
        CLA
        SAS I OVERFLOW  /WAS OVERFLOW CHECK REQUESTED
        JMP FASDAT      /DATA ERROR
        RIR 2S
        SPI I
        JMP FASDAT      /DATA ERROR=NOT OVERFLOW
        CLC
        DAC I CHIXE
        LIO I CHNFLG
        LAC ERSTART     /CHECK IF IN PROCESS OF PROCESSING AN ERROR
        SZA
        LIO SCHFLG      /YES=GET FLAGS FROM SAVED REGISTER
        LFI
        DZM ERSTART
        JMP I OVFX

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/TAPE PROBLEMS

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ATAPE,  RRI 200
        RIR 9S
        LAW 2000
        SPI
        JMP 1TERR
        RRI 1200        /WAS IT AN EOT
        AND (-0
        RIR 9S
        LAW 1000        /EOT ?
        SPI
        JMP 1TERR
        JSP 1HLT        /ERROR (?????)

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FASERR, REPEAT 4,0 /DATA ON LAST FASTRAND ERROR

FASDAT, IDX FASERR
 RRI 100
 DIO FASERR+1
 RRI+1000
 DIO FASERR+2
 RRI+1100
 DIO FASERR+3

/DATA ERRORS=ENTRY ALSO FOR DATA LATE

DATA, CLA /WAS THERE PREVIOUS DATA ERROR

SAD ERSTART

JMP DATA1

~~CLA /WAS THIS CORRECTIVE MEASURE~~

~~SAS ERGO~~

~~JSP 1HLT /WHAT CAN WE DO~~

ISP ERRCNT

JMP DATA2 /NOT COMPLETED THE COUNT YET

CLC

DAC I CHIXE /MARK AS NOT=HANDLED ERROR

JSP 1HLT /NON=RECOVERABLE ERROR

DATA1, LIO I CHNFLG /SAVE FLAGS

DIO SCHFLG

LIO I CHIX

DIO SCHIX /SAVE DEBREAK REGISTER

LAW I 4

DAC ERRCNT

IDX ERSTART /MARK ERSTART NOT ZERO=I,E,PROCESSING AN ERROR

DATA2, RRI /TAPE OR FASTRAND

LAI

AND (3

SAS (1

HLT

/TAPE ERROR = CAN'T GET HERE!!(??)

RRI 500
 LAI
 AND (177777
 IOR (140000
 DAC RED02
 SUB (1
 DAC RED01

/RELOCATE

RED04,

LAC I RED02
 SZA I
 JMP RED03
 AND (700000
 SAD (400000
 JMP RED03
 IDX RED02
 JMP RED04

/LOOK FOR 0 OR ADD

/FOUND HALT

/FOUND JMP

RED03,

LAC (RED0H
 DAC RED05
 CLF 1

/SET POINTER TO COPY BACKWARDS

RED08,

LIO I RED02
 DIO I RED05
 LAI
 AND (700000
 SAD (500000
 JMP RED06
 SAS (100000

/MOVE COMMAND

SZA I
 JMP RED07

/WAS IT A WORD COUNT COMMAND

RED09,

LAW I 1
 ADD RED05
 DAC RED05
 LAW I 1
 ADD RED02
 DAC RED02
 SUB RED01
 SPA
 STF 1
 JMP RED08

/DECREMENT COUNTS

/IF STRICTLY BEFORE WORD BEFORE ERROR SET FLAG

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RED07,    LAI
          RIL 58
          SPI I      /DID WE REQUEST AN ERROR
          JMP RED09   /NO
          AND (-10000
          SZF 1        /DO WE WANT TO REMOVE BREAKS
          DAC I RED05  /REMOVE BREAK BIT
          JMP RED09

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RED06,    LAC RED05   /MAKE INTO A PHYSICAL ADDRESS
          AND (37777

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TDATA2, DAC RED05
RED010, JSP I GO    /RE-EXECUTE THE COMMANDS
RED05,    0

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          LIO SCHFLG
          LFI
          DZM ERSTART
          DZM ERGO
          CLC
          DAC I CH1X
          JMP I SCH1X

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ERGO,    0

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REDOC,    REPEAT 103,,0
RED0H:    0

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ERSTART,  0
ERRCNT,   0
SCH1X,    0
SCHFLG,   0
RED02,    0
RED01,    0

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CON,      CONSTANTS
FOO,      FLEXO FOO

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/TABLES FOR IOP III

16224/ TTBL:
 16232/ FTBL:
 16400/ BITTB1,
 16500/ BITTB2,
 16600/ BITTB3,
 16700/ BITTB4,

17000/
 FRELST: REPEAT 4,240000
 REPEAT 174,200001
 REPEAT 4,300000
 REPEAT 164,200001
 REPEAT 10,400000
 REPEAT 4,240000
 REPEAT 174,200001

/MARK FIRST THIRD HELD BY USER 1
 /MARK FIRST TRACK HELD BY SWAPPER

17600/
 HLDHT0, DCH I .
 REPEAT 15.,0
 FREBL0, 400000
 REPEAT 15.,-0

/CORE 17 ADDRESSES TO ADDRESS

157700/
 TUSER, 0
 DCONT, 0
 WAIT4, 0
 SWAP, 0
 ,SETUP, 0
 RPC, 0
 SCORE, 0
 GO, 0
 WAIT, 0
 CHIX, 0
 CHNFLG, 0
 CHIXE, 0
 16USER, 0
 RELEASE, 0
 CH1A, 0
 COMPX, 0
 OVFX, 0
 OVERFLOW, 0
 THIRD, 0
 READ, 0
 READ+1, 0
 WAIT2, 0
 1EMPTY, 0
 FILLSPC, 0
 1HLTF, 0
 TIME, 0
 C16FIN, 0
 C16FOU, 0
 C16FSW, 0
 SWAPE5, 0

PAGE 12

/CORE 4,0 REFERENCES

ERCOD1,	IOR 102
ERCOD2,	IOR 103
RERCD1,	102
RERCD2,	103
AC,	JDA 7775
PC,	IOR 35
IO,	JDA 7776
FLAGS,	JDA 7777
IOPRT,	IOR 4
IOPRT1,	IOR 5
TRAPPC,	IOR 77
RTRAPC,	77
IOPMAX,	IOR 74
BOUND,	36
PRIORI,	IOR 15

IOPTSU=106

START HLT=JMP